

Software Reverse Engineering Course

- Site: OnHexGroup.ir
- Youtube: @onhexgroup
- Telegram: onhex_ir
- Twitter: @onhexgroup
- Github: onhexgroup

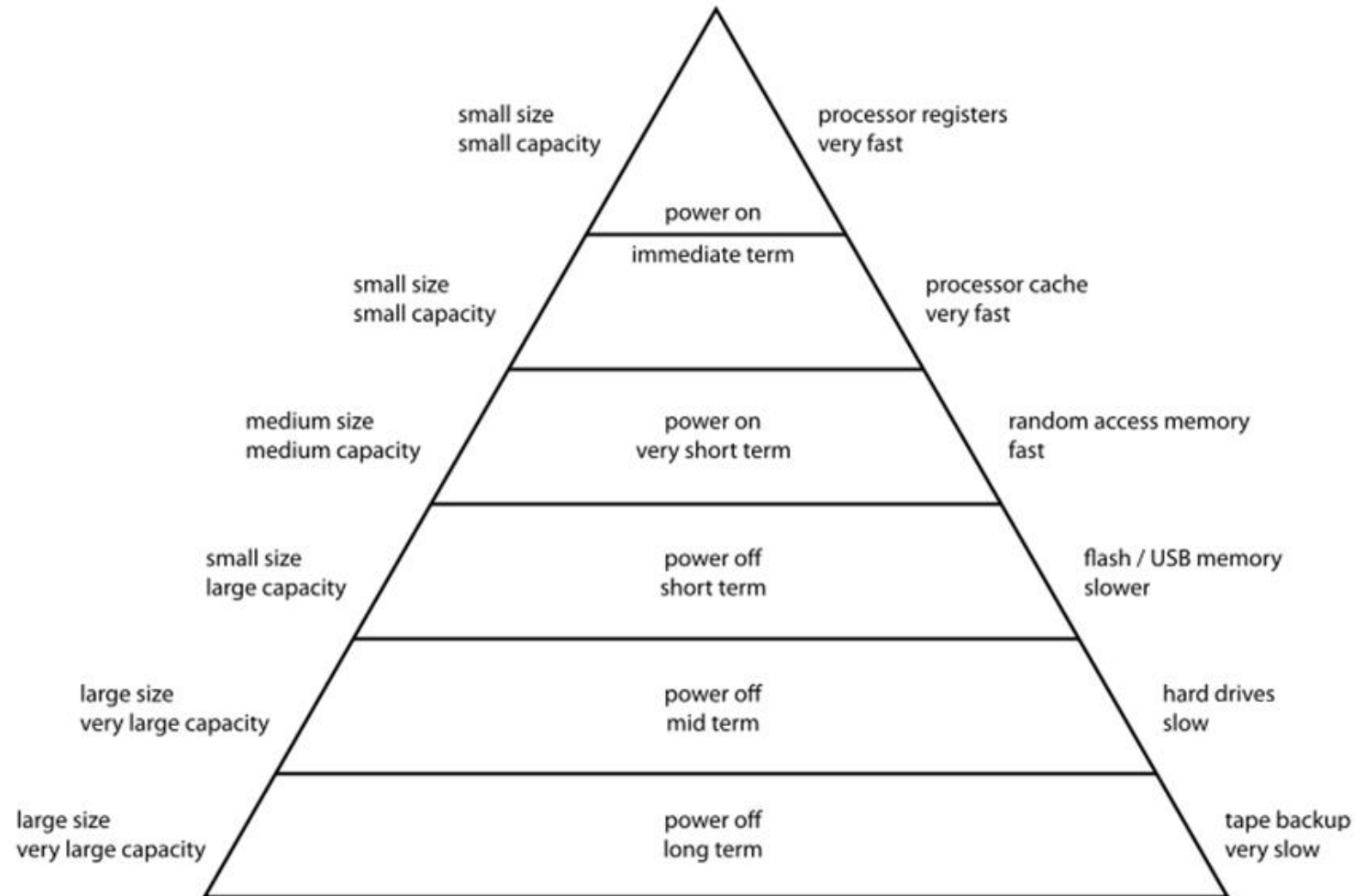
Windows
(x86 / x64)

By: OnhexGroup

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Memory Hierarchy

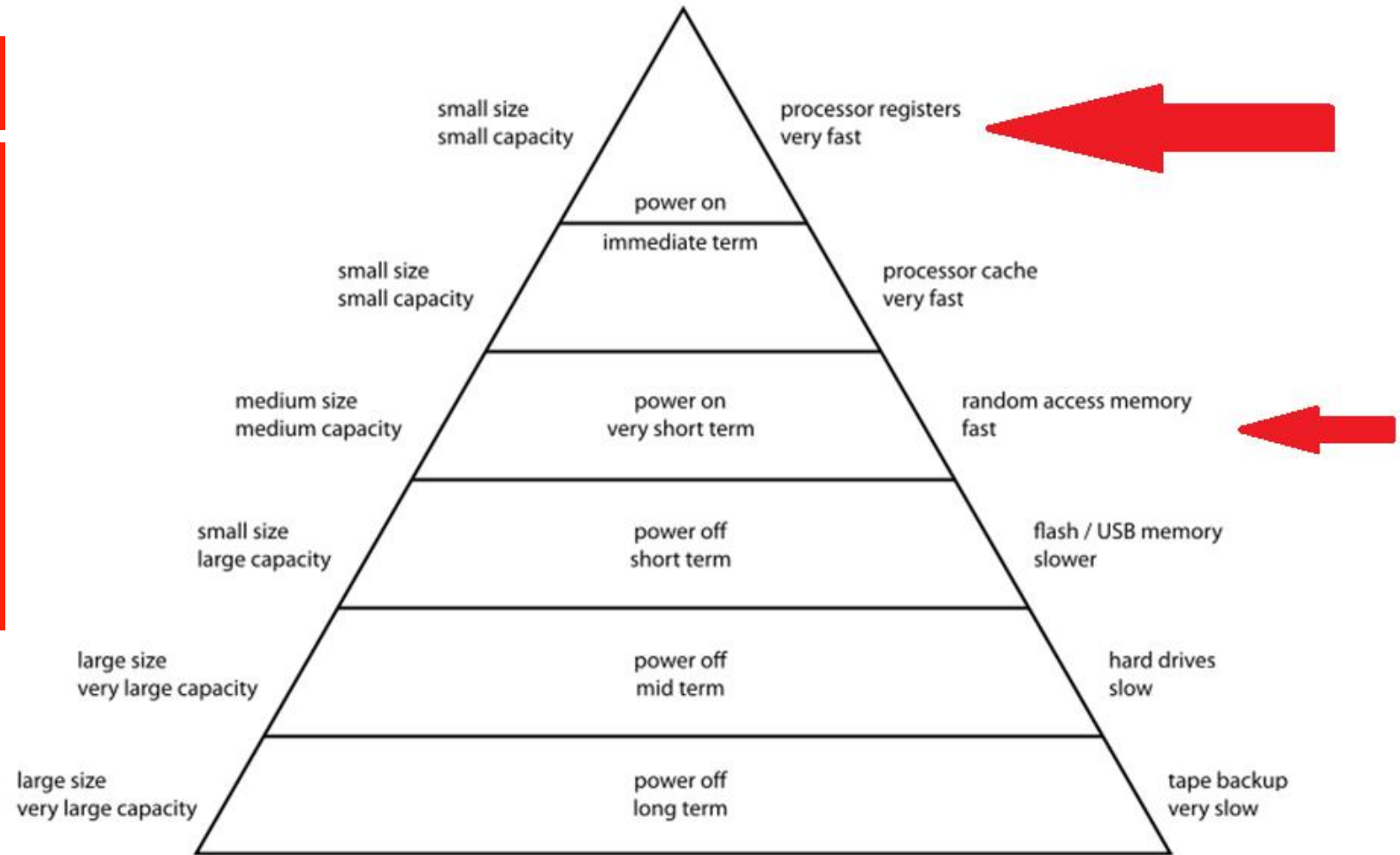
Computer Memory Hierarchy



Telegram: onhex_ir

Memory Hierarchy

Computer Memory Hierarchy



Youtube: @onhexgroup

Measurement Units

Description	Type
Smallest unit. Value is either 0 or 1	Bit

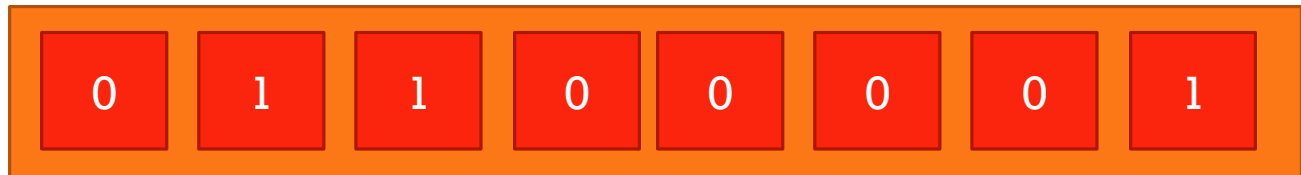
0

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Github: onhexgroup

Measurement Units

Description	Type
Smallest unit. Value is either 0 or 1	Bit
Every 8 bits equal 1 byte	Byte



Github: onhexgroup

Addressing

- **Smallest addressable unit: Byte**
 - **32bit Systems = Address space: 2^{32} bytes**
 - $2^{32} = 4,294,967,296$
 - Hex representation: **0xFFFFFFFF**
 - **64-bit Systems = Address space: 2^{64} bytes**
 - $2^{64} = 18,446,744,073,709,552,000$
 - Hex representation: **0xFFFFFFFFFFFFFFFF**

Youtube: @onhexgroup

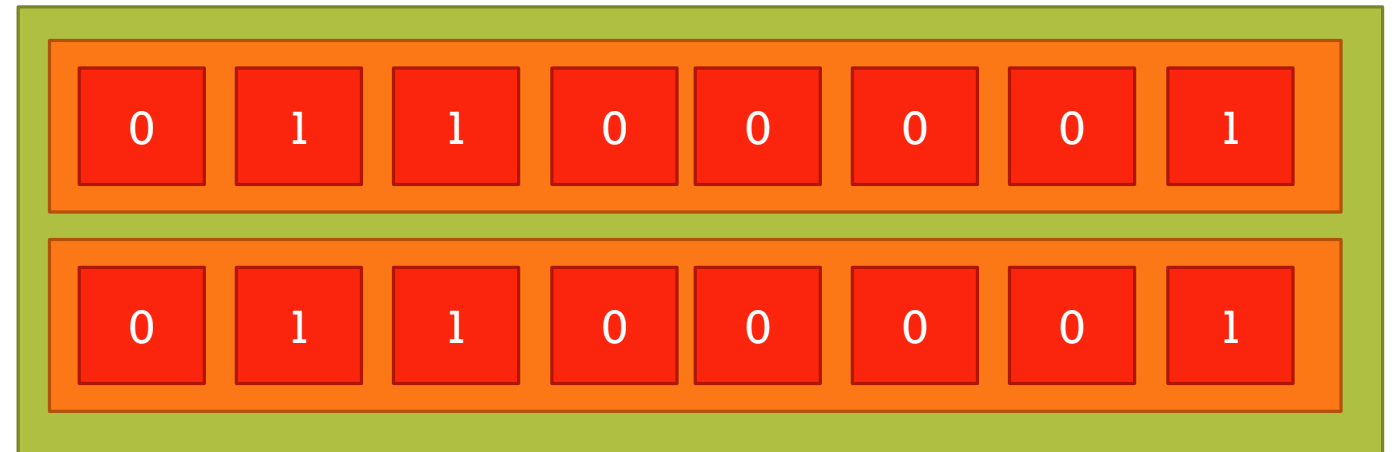
ASCII Code

Control Characters				Graphic Symbols											
Name	Dec	Binary	Hex	Symbol	Dec	Binary	Hex	Symbol	Dec	Binary	Hex	Symbol	Dec	Binary	Hex
NUL	0	0000000	00	space	32	0100000	20	@	64	1000000	40	'	96	1100000	60
SOH	1	0000001	01	!	33	0100001	21	A	65	1000001	41	a	97	1100001	61
STX	2	0000010	02	"	34	0100010	22	B	66	1000010	42	b	98	1100010	62
ETX	3	0000011	03	#	35	0100011	23	C	67	1000011	43	c	99	1100011	63
EOT	4	0000100	04	\$	36	0100100	24	D	68	1000100	44	d	100	1100100	64
ENQ	5	0000101	05	%	37	0100101	25	E	69	1000101	45	e	101	1100101	65
ACK	6	0000110	06	&	38	0100110	26	F	70	1000110	46	f	102	1100110	66
BEL	7	0000111	07	'	39	0100111	27	G	71	1000111	47	g	103	1100111	67
BS	8	0001000	08	(	40	0101000	28	H	72	1001000	48	h	104	1101000	68
HT	9	0001001	09	)	41	0101001	29	I	73	1001001	49	i	105	1101001	69
LF	10	0001010	0A	*	42	0101010	2A	J	74	1001010	4A	j	106	1101010	6A
VT	11	0001011	0B	+	43	0101011	2B	K	75	1001011	4B	k	107	1101011	6B
FF	12	0001100	0C	,	44	0101100	2C	L	76	1001100	4C	l	108	1101100	6C
CR	13	0001101	0D	-	45	0101101	2D	M	77	1001101	4D	m	109	1101101	6D
SO	14	0001110	0E	.	46	0101110	2E	N	78	1001110	4E	n	110	1101110	6E
SI	15	0001111	0F	/	47	0101111	2F	O	79	1001111	4F	o	111	1101111	6F
DLE	16	0010000	10	0	48	0110000	30	P	80	1010000	50	p	112	1110000	70
DC1	17	0010001	11	1	49	0110001	31	Q	81	1010001	51	q	113	1110001	71
DC2	18	0010010	12	2	50	0110010	32	R	82	1010010	52	r	114	1110010	72
DC3	19	0010011	13	3	51	0110011	33	S	83	1010011	53	s	115	1110011	73
DC4	20	0010100	14	4	52	0110100	34	T	84	1010100	54	t	116	1110100	74
NAK	21	0010101	15	5	53	0110101	35	U	85	1010101	55	u	117	1110101	75
SYN	22	0010110	16	6	54	0110110	36	V	86	1010110	56	v	118	1110110	76
ETB	23	0010111	17	7	55	0110111	37	W	87	1010111	57	w	119	1110111	77
CAN	24	0011000	18	8	56	0111000	38	X	88	1011000	58	x	120	1111000	78
EM	25	0011001	19	9	57	0111001	39	Y	89	1011001	59	y	121	1111001	79
SUB	26	0011010	1A	:	58	0111010	3A	Z	90	1011010	5A	z	122	1111010	7A
ESC	27	0011011	1B	;	59	0111011	3B	[	91	1011011	5B	{	123	1111011	7B
FS	28	0011100	1C	<	60	0111100	3C	\	92	1011100	5C		124	1111100	7C
GS	29	0011101	1D	=	61	0111101	3D	]	93	1011101	5D	}	125	1111101	7D
RS	30	0011110	1E	>	62	0111110	3E	^	94	1011110	5E	~	126	1111110	7E
US	31	0011111	1F	?	63	0111111	3F	_	95	1011111	5F	Del	127	1111111	7F

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Measurement Units

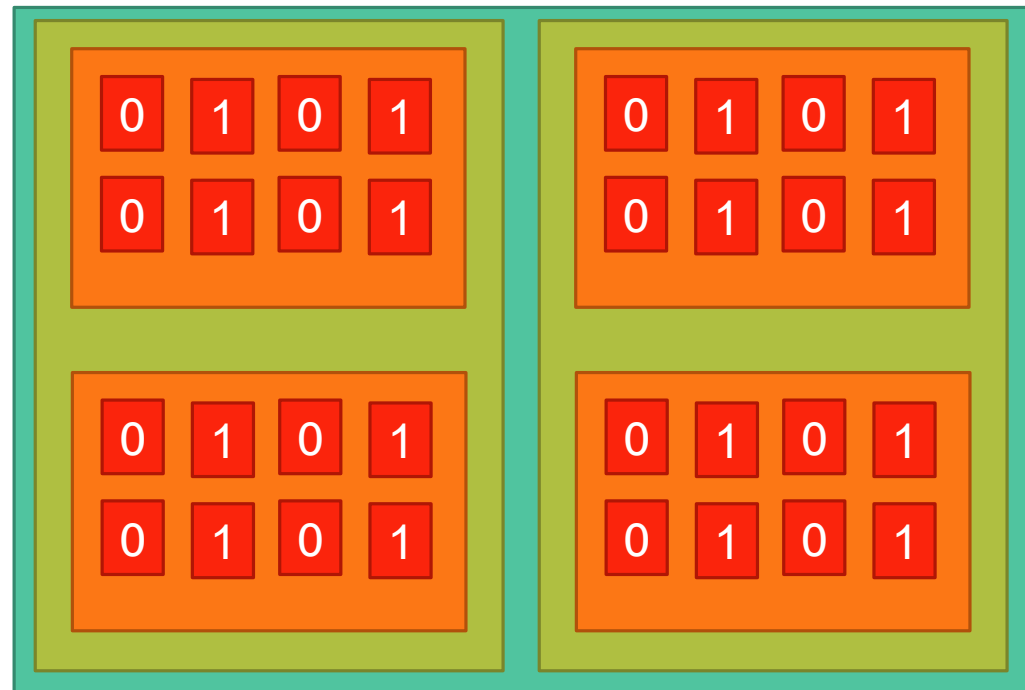
Description	Type
Smallest unit. Value is either 0 or 1	Bit
Every 8 bits equal 1 byte	Byte
Every 16 bits or 2 bytes is a word.	Word



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Measurement Units

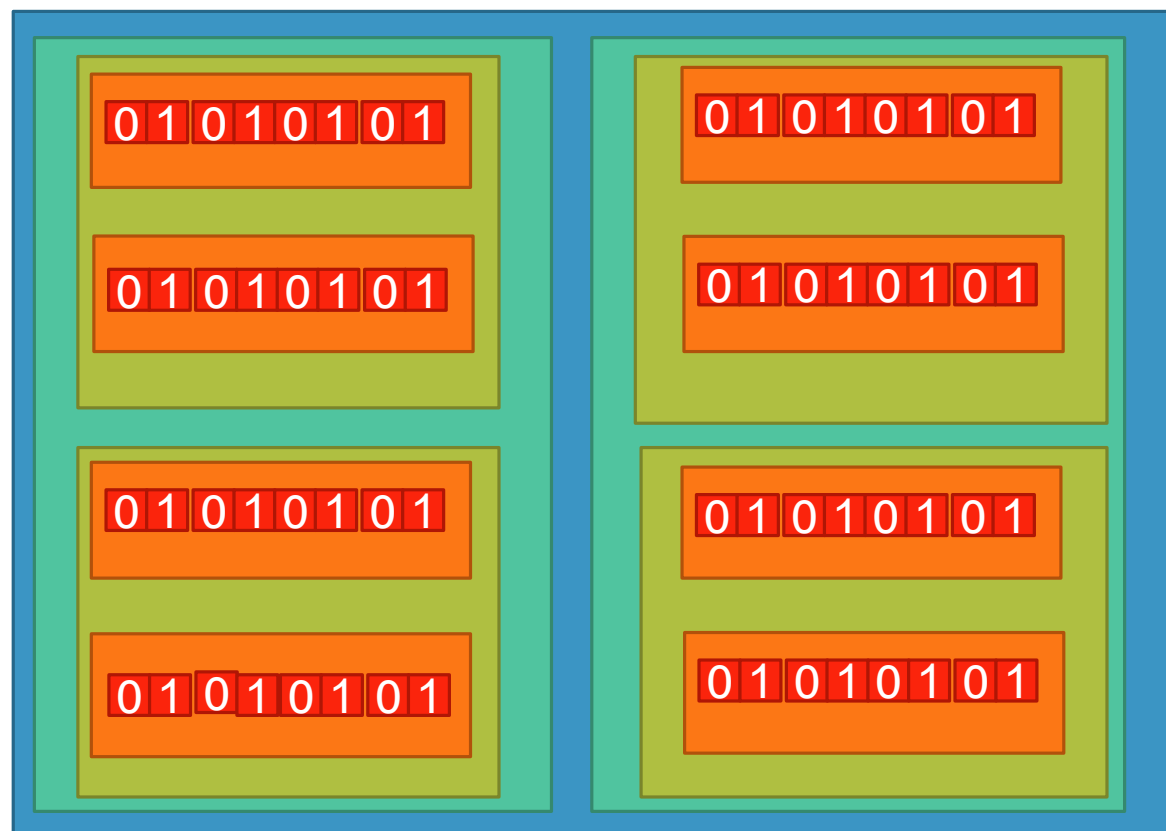
Description	Type
Smallest unit. Value is either 0 or 1	Bit
Every 8 bits equal 1 byte	Byte
Every 16 bits or 2 bytes is a word.	Word
Every 32 bits or 4 bytes or 2 words is equal to a Dword.	DWord



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Measurement Units

Description	Type
Smallest unit. Value is either 0 or 1	Bit
Every 8 bits equal 1 byte	Byte
Every 16 bits or 2 bytes is a word.	Word
Every 32 bits or 4 bytes or 2 words is equal to a Dword.	DWord
Every 64 bits or 8 bytes or 4 words or two Dwords is equal to one Qword.	QWord



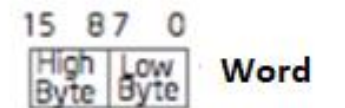
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Data Types in C Language

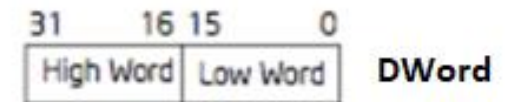
In C: char



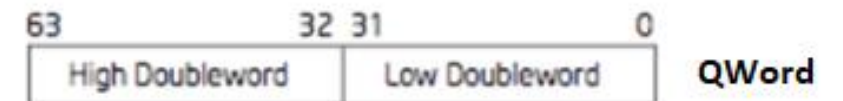
In C: short



In C: int/long



In C: double/long long



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Data Types

Description	Type
Smallest unit. Value is 0 or 1	Bit
Known as DB in Assembly	Byte
Known as DW in Assembly	Word
Known as DD in Assembly	DWord
Known as DQ in Assembly	QWord

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Data Types

Onhex_ir

Char

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Data Types

o

n

h

e

x

_

i

r

Data Types

Char

Ascii-binary

o

01101111

n

01101110

h

01101000

e

01100101

x

01111000

-

01011111

i

01101001

r

01110010

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Data Types

Char

Ascii-binary

Ascii-Hex

o

01101111

6F

n

01101110

6E

h

01101000

68

e

01100101

65

x

01111000

78

_

01011111

5F

i

01101001

69

r

01110010

72

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Data Types

Char

Ascii-binary

Ascii-Hex

Byte

o

01101111

6F

6F

n

01101110

6E

6E

h

01101000

68

68

e

01100101

65

65

x

01111000

78

78

_

01011111

5F

5F

i

01101001

69

69

r

01110010

72

72

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Data Types

Onhex_ir

Char

Byte

Word

o

6F

6F

6E

o

n

n

6E

h

68

68

65

h

e

e

65

x

78

78

5F

x

_

_

5F

i

69

69

72

i

r

r

72

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Data Types

Char

Byte

Word

DWord

o

6F

6F

6E

n

6E

6F

6E

68

65

o

n

h

e

h

68

68

65

e

65

x

78

78

5F

_

5F

78

5F

69

72

x

_

i

r

i

69

69

72

r

72

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Endianness

- **Horizontal data layout in an addressable memory space**
- **Types**
 - **Big-Endian**
 - **Little-Endian**

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Endianness

1402



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Endianness

1402



1402



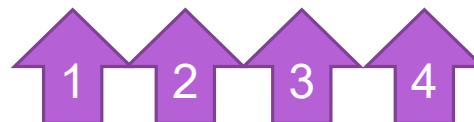
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Endianness

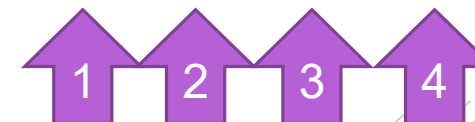
1402



1402



1402

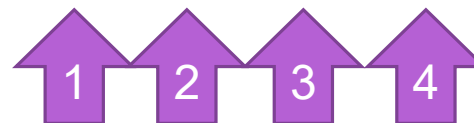


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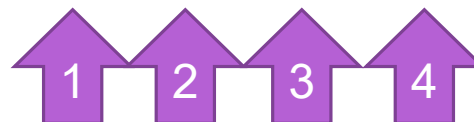
Endianness

Big Endian

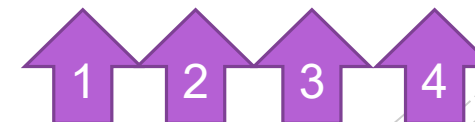
1402



1402

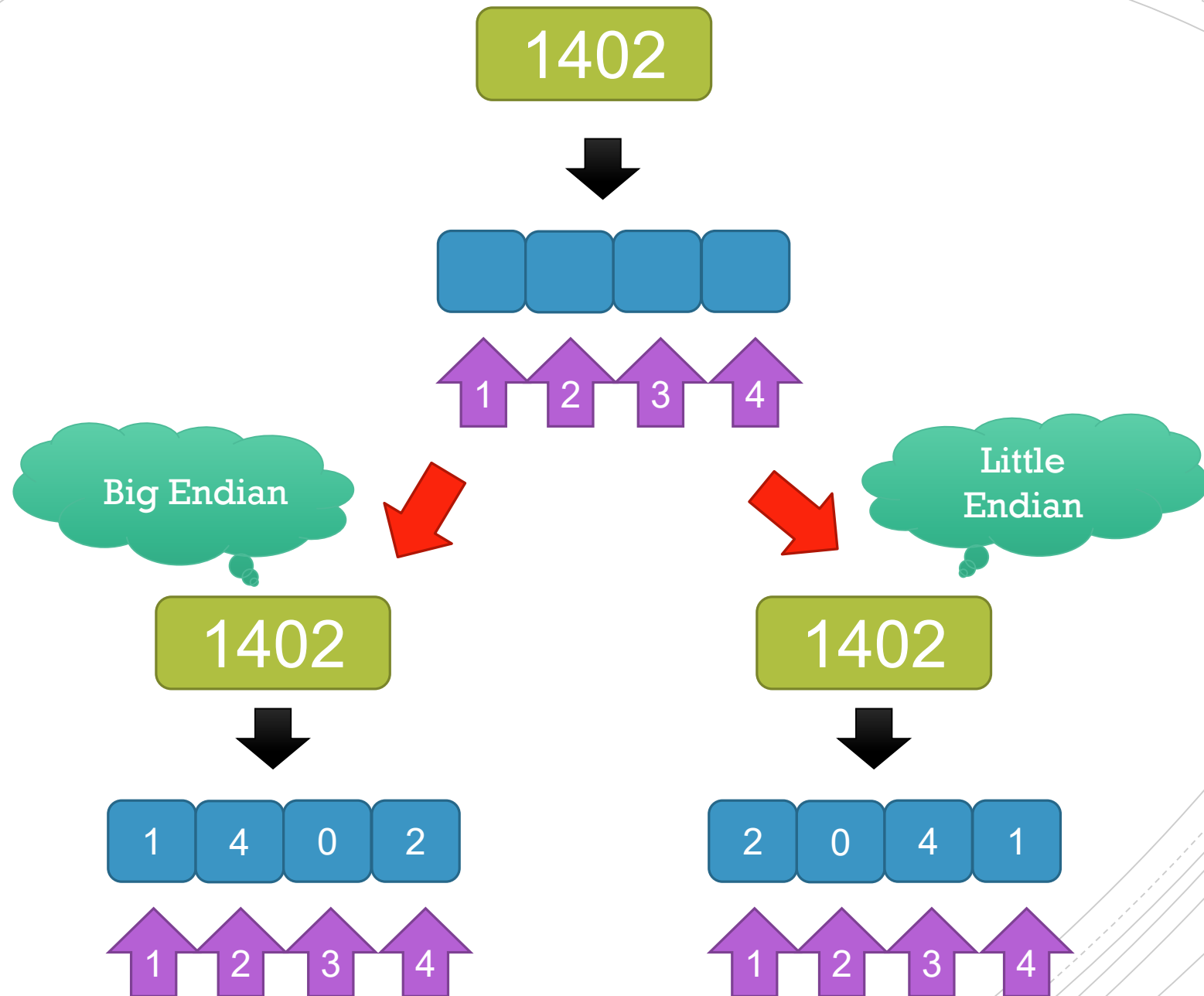


1402



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Endianness



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Endianness

Data
0x01020304

Little Endian

Address	0x100	0x101	0x102	0x103
	0x04	0x03	0x02	0x01

Big Endian

Address	0x100	0x101	0x102	0x103
	0x01	0x02	0x03	0x04

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Endianness

- **x86 architecture is Little-Endian, Network traffic is usually Big-Endian**
- **Endianness applies to memory, not registers**
- **Endianness applies to bytes, not bits**

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Course Foundation

- **In this course, we cover:**
- x86 processor family 32-bit and 64-bit architectures
- Protected Mode
- Reverse Engineering of compiled programs (C / C++)
- **Memory representation will be Little-Endian**

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Char

Byte

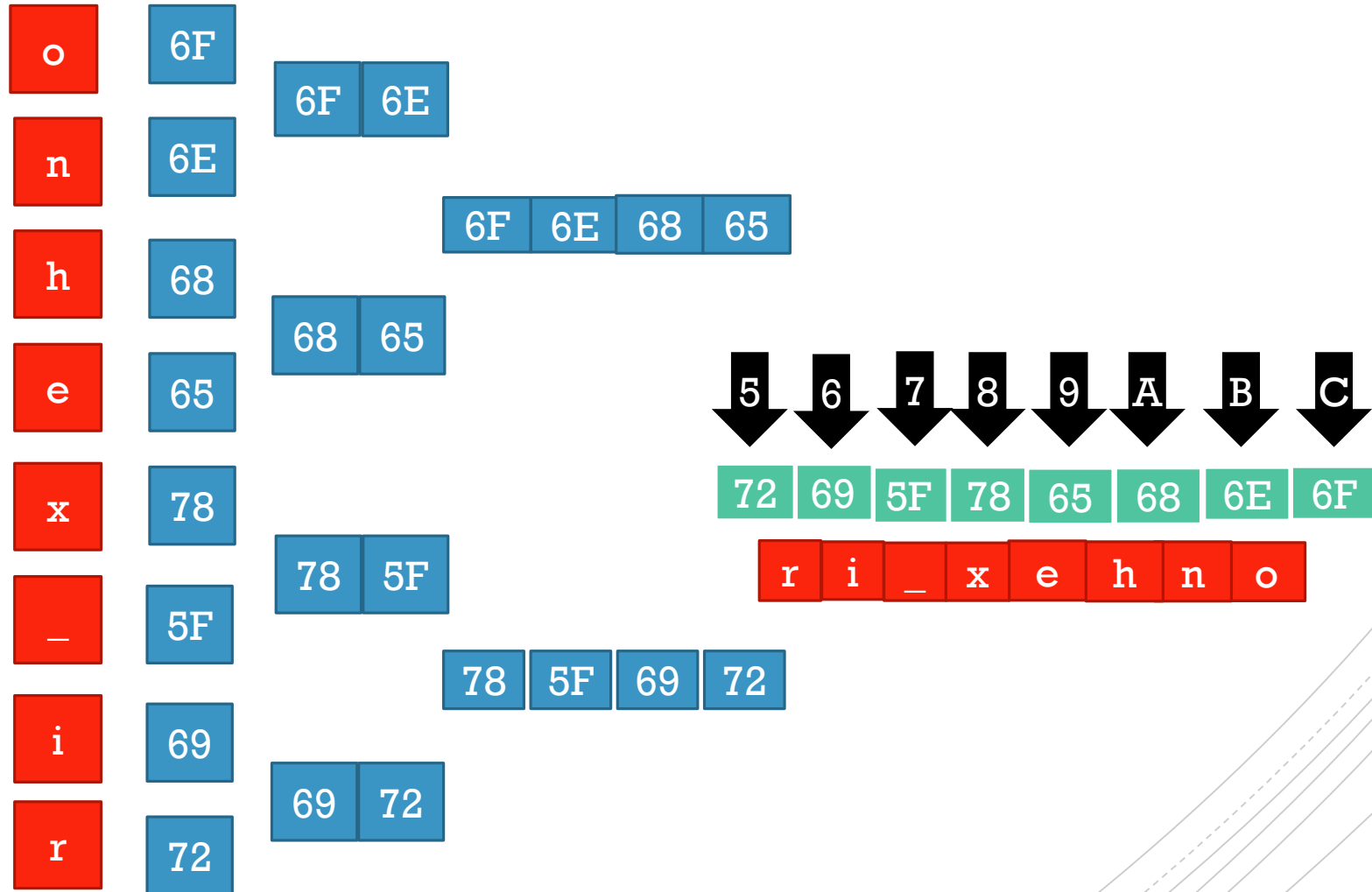
Word

DWord

QWord

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Example



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References

- [https://en.wikipedia.org/wiki/Endianne SS](https://en.wikipedia.org/wiki/Endianne_SS)
- [https://en.wikipedia.org/wiki/Memory hierarchy](https://en.wikipedia.org/wiki/Memory_hierarchy)
- <https://www.geeksforgeeks.org/memory-hierarchy-design-and-its-characteristics/>